

Lista de Exercícios: Octal para Binário

Instruções

Converta os números octais abaixo para o sistema binário.

Exercícios	Gabarito
1. $5_8 =$	1. $5_8 = 101_2$
2. $6_8 =$	2. $6_8 = 110_2$
3. $10_8 =$	3. $10_8 = 1000_2$
4. $12_8 =$	4. $12_8 = 1010_2$
5. $15_8 =$	5. $15_8 = 1101_2$
6. $20_8 =$	6. $20_8 = 10000_2$
7. $25_8 =$	7. $25_8 = 10101_2$
8. $31_8 =$	8. $31_8 = 11001_2$
9. $36_8 =$	9. $36_8 = 11110_2$
10. $45_8 =$	10. $45_8 = 100101_2$
11. $52_8 =$	11. $52_8 = 101010_2$
12. $62_8 =$	12. $62_8 = 110010_2$
13. $77_8 =$	13. $77_8 = 111111_2$
14. $100_8 =$	14. $100_8 = 1000000_2$
15. $113_8 =$	15. $113_8 = 1001011_2$
16. $120_8 =$	16. $120_8 = 1010000_2$
17. $140_8 =$	17. $140_8 = 1100000_2$
18. $144_8 =$	18. $144_8 = 1100100_2$
19. $175_8 =$	19. $175_8 = 1111101_2$
20. $226_8 =$	20. $226_8 = 10010110_2$

Exemplo Resolvido

Converter 226_8 para binário:

Tabela de conversão

Binário	Octal
0	0
1	1
10	2
11	3
100	4
101	5
110	6
111	7

1. Substituir cada algarismo octal

$$2_8 = 010_2$$

$$2_8 = 010_2$$

$$6_8 = 110_2$$

2. Juntar os grupos

$$226_8 = 010\ 010\ 110_2$$

3. Remover o zero inicial

$$226_8 = 10010110_2$$